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Midterm

ITAI-3377 A.I at the Edge

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AI-Integrated Industrial Iot Powered Damage Control System for Naval Vessels

A modern warship is a floating town that makes its own electricity. When a power cable is damaged or a pump motor shorts bad things happen fast. Lights go out, radar shuts down, even fire fighting gear loses force. Installing low cost sensors on every breaker and pump then letting an onboard AI react in milliseconds, keeps the lights on and the crew safe. The catch: thousands of new network connected devices are now attractive to hackers! This plan shows how to add intelligence and keep attackers out.

1. How the AI-Integrated Iot Fits Together

- **Smart Breakers and Pumps**

What are they? - Electronic circuit breakers and pump controllers that measure current, temperature, vibration, and send those numbers every tenth second.

Why untrusted? - Anyone in the space could plug a laptop into their service port.

- **Control Gateways (PLCs)**

PLCs = Programmable Logic Controllers. Rugged computers wired right above the breakers. They translate the old serial messages into encrypted Ethernet messages and still hold the classic “trip the breaker if current spikes” rule.

- **Edge Data Center**

Two suitcase sized racks with graphics processors run three pieces of software:

- a **load shedding AI** that decides which non essential circuit to turn off first,
- a **thermal camera AI** that spots a hot bulkhead before fire starts,
- a **traffic anomaly AI** that notices strange network behaviour.

- **Ship Network**

One fibre ring moves data around the ship. It is split into four “virtual lanes”: Power gear, Other machinery, Office computers, and Crew Wi-Fi. Encryption on every link stops eavesdropping; certificates on every device stop strangers from joining.

- **Pier Side Cloud Connection**

When the ship is in port, an encrypted satellite link pulls in firmware updates and new AI models that were tested ashore. Keys used on this link expire in 24 hours, so they can’t be replayed later at sea.

2. Weaknesses Discovered

ID	Description	Result if Exploited
V1	Breaker accepts any USB firmware file	Breaker fails to open -> electrical fire
V2	Sensor hub still “admin/admin”	Hacker jumps into control network
V3	Crew Wi-Fi cable mislabelled	Phone malware reaches safety gear
V4	Old Modbus messages are unencrypted	Replayed frame trips vital breaker
V5	Load shedding AI trusts any sensor data	Fake overload causes black out
V6	Control room web page uses outdated library	Phishing steals override token
V7	Update tool pins a soon expiring certificate	Crew may install fake updates
V8	Badge alone opens the control console	Lost badge silences alarms

3. Defence Strategy

3.1. Secure Design Principles

Breakers now boot only after verifying a digital signature stored in a factory burned security chip. If someone loads a rogue file (V1), the breaker refuses to power up; the fault is obvious instead of hidden. Similar signature checks guard every software container on the server before it starts.

3.2. Authentication & Access Control

Every network device; breakers, sensors, laptops must show a short lived digital certificate to join the ship’s network. Control room consoles ask for both an NFC badge and a rotating six-digit PIN, so losing the badge alone (V8) is harmless. Default passwords on sensor hubs are replaced by random values stored in an onboard password vault (V2).

3.3. Encryption & Data Protection

Legacy serial messages are wrapped inside TLS encrypted packets the instant they reach the first gateway and each packet carries a running number. Old or replayed messages (V4) are ignored automatically.

3.4. Network Security

The single fibre ring is divided into four virtual lanes. A switch that ever sees Crew Wi-Fi traffic on the Power lane quarantines that port within seconds (V3). All links use MACsec blocking eavesdropping.

3.5. Secure Software Development

The control room web page is rebuilt every Friday. An automated scan rejects any library with a known vulnerability, and strict browser rules block inline scripts. Even if a future phish lands (V6) the payload cannot run.

3.6. Physical Security

USB service ports on breakers are sealed after firmware loading. Locked covers require a signed work chit to reopen, reducing the chance of casual tampering (V1). Drone charging cabinets are inside a keyed space; removing a drone triggers a camera snapshot.

3.7. Security Monitoring & Incident Response

A network analytics program (Zeek) mirrors every packet. If a breaker opens without a matching “trip reason” tag or if an untrusted device appears, an automated playbook both alerts the repair party and blocks the offender’s address. Mean time to contain stays under five minutes in drill results.

3.8. Specific Measures for AI Models & Data

A 20 line guardian rule set in the PLC checks real watt meter readings before acting on the AI’s shed command. Adversarial sensor data (V5) therefore fails the check. Model files are stored in a ‘read only’, signed repository; hash verification at load time blocks silent model swaps. New adversarial examples discovered during drills are folded into quarterly retraining.

4. Implementation Plan

Week	Task	Lead Role	Tools
1-2	Inventory devices and run baseline scan	Security Engineer	Plscan, Nmap
3-4	Install signed breaker firmware, seal USB ports	Electrician’s Mate	Fwupd, tamper seals
5-6	Roll out device certificates	Network Team	Cisco ISE, Ansible
7-8	Insert TLS gateways on serial links	OT Engineer	HMS Anybus
9	Load guardian rules; perform adversarial retraining	Controls Engineer, Data Scientist	PLC ladder editor TensorFlow
10	Conduct cyber drill	Student Red & Blue teams	CALDERA, Wi-Fi test kit
11-15	Fix issues, tune alerts	Same teams	SIEM console
16	Final readiness brief and document hand-off	Project Lead	Risk scorecard

5. Penetration Testing Simulation

- **Rotation of roles.** Each student spends half a day as an attacker and half a day as defender.

- **Scenario 1 – USB attack (V1).** Attacker loads unsigned firmware; breaker refuses to boot, alarm sounds.
- **Scenario 2 – Evil twin Wi-Fi (V3).** Fake access point fails certificate check; switch quarantines port.
- **Scenario 3 – Sensor spoof (V5).** Script feeds high load numbers; guardian rules block AI order; SIEM logs incidents.

Assessment & Improvement

After every scenario the group records:

- detection time,
- containment time,
- what stopped the attack,
- what alert felt noisy or missing.

Findings feed the Week 11 tuning sprint. Alert thresholds adjusted, guardian rules expanded, and any long lead time fixes added to the ship's change control log.

6. Conclusion

Digitising the ship's power plant with smart breakers, cameras, and onboard AI delivers two priceless gains: speed (milliseconds instead of minutes to isolate a fault) and foresight (thermal and traffic anomalies spotted before damage spreads). The same sensors however, widen the attack surface. The security design presented here keeps risk in check by:

- starting with hardware rooted signatures on every executable part,
- enforcing strong certificate based identity on every wire,
- encrypting and segregating all data in motion,
- placing an old fashioned deterministic “guardian” between AI advice and physical action,
- tying continuous monitoring to an automated response that a repair party can trust.

A sixteen-week rollout schedule shows the defences can be phased in with existing crew skills and COTS tools, while the red/blue team drills prove each layer works under pressure. Follow on work; future proofing certificates, tuning alerts for noise, expanding adversarial training which already has a slot in the change log so the protection evolves alongside the tech.

The result is a smarter, faster, and measurably safer ship that is ready to keep fighting when the next casualty alarm rings.

References

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